Sensor Data Logger

Gaucho Rocket Project - 26-27

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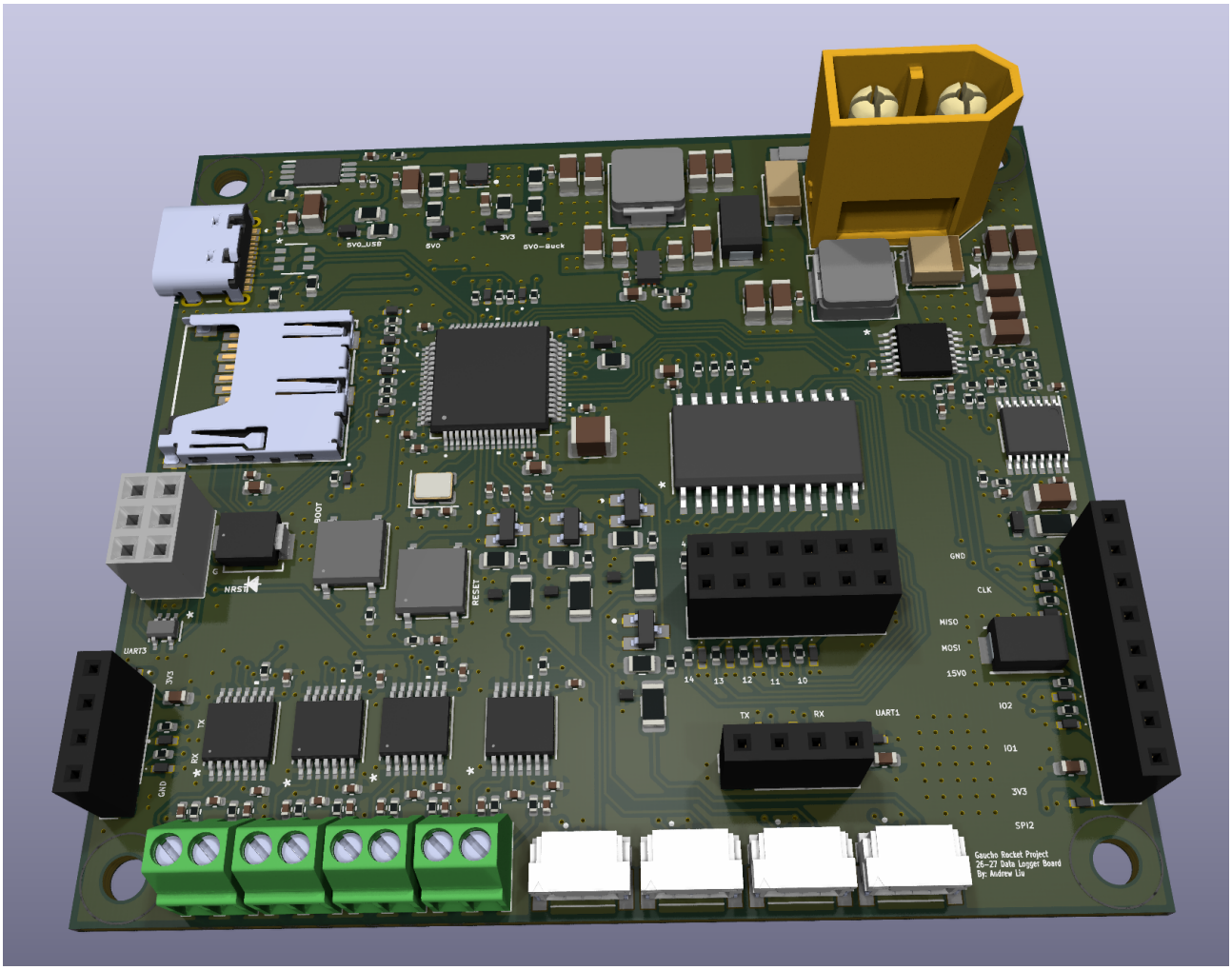


Figure 1: Board 3D Viewer on KiCad

1 Introduction

In the past we have been purchasing micro-controller developer kits, amplifier breakout boards, and passive components and creating a perfboard as our sensor logging system. A full writeup of the previous system is available here.

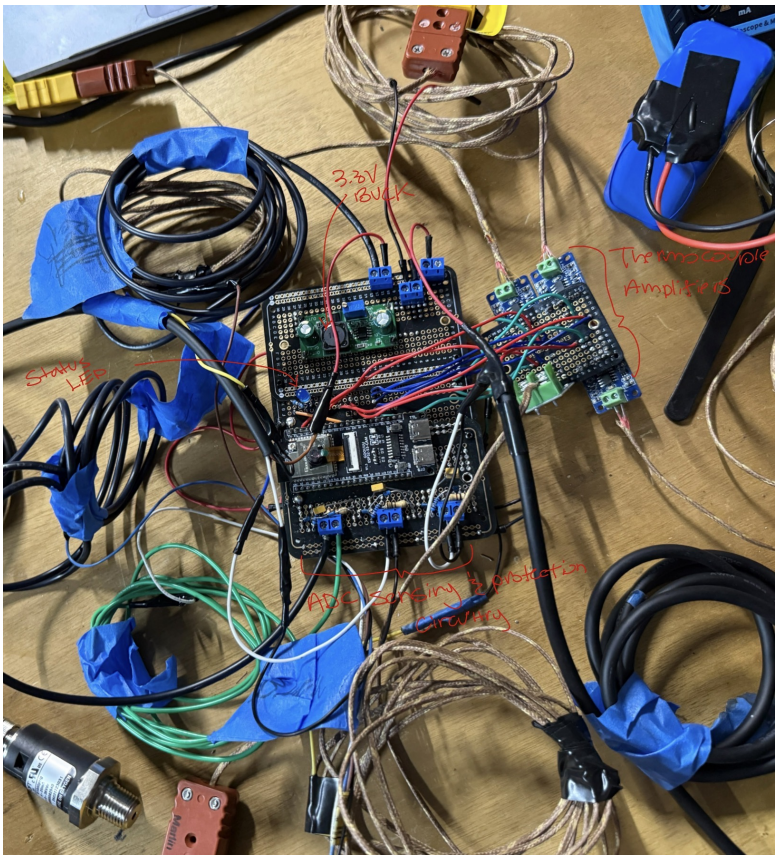


Figure 2: Previous perfboard

I want to step away from this, because a perfboard brings in many points of failure including: exposed traces increasing chances of shorts, loose connections due to screw terminals and bad solder joints, and board being fragile during transport.

Due to rocket space and weight constraints, I will break up the avionics PCB system into three separate boards, to allow for flexibility for future and current rockets.

1. Sensor Logger Board (covered in this writeup)
2. Flight Telemetry Board
3. Extra Peripherals Board (e.g., extra controls and sensors, custom for each rocket)

2 Board Specifications

Microcontroller	STM32L476RGT6
Power input	9V-12.6V LIPO
Power rails	15V, 5V, 3.3V
Temperature inputs	MAX31856 Thermocouple Amplifier Chips
Pressure inputs	4-20mA through 130 Ohm Shunt Resistor to ADC
Data storage	MicroSD (SDMMC)
Interfaces	SPI, UART x2
Debug	USB, SWD
PCB	4-layer, 78.68 mm * 71.02 mm
Status	Ordered

Table 1: Key specifications.

3 System Architecture

3.1 Power Path

In the past, when testing the pressure transducer, we noticed that when hot-plugging the battery, the source voltage spikes to the output for about 100 microseconds, frying our ESP32 boards.



Figure 3: Transient voltage spike on scope

In order to combat this, I made sure to power the pressure transducers using regulated rails and added TVS and voltage clamping protection circuitry at the ADC pins. The power architecture goes like:

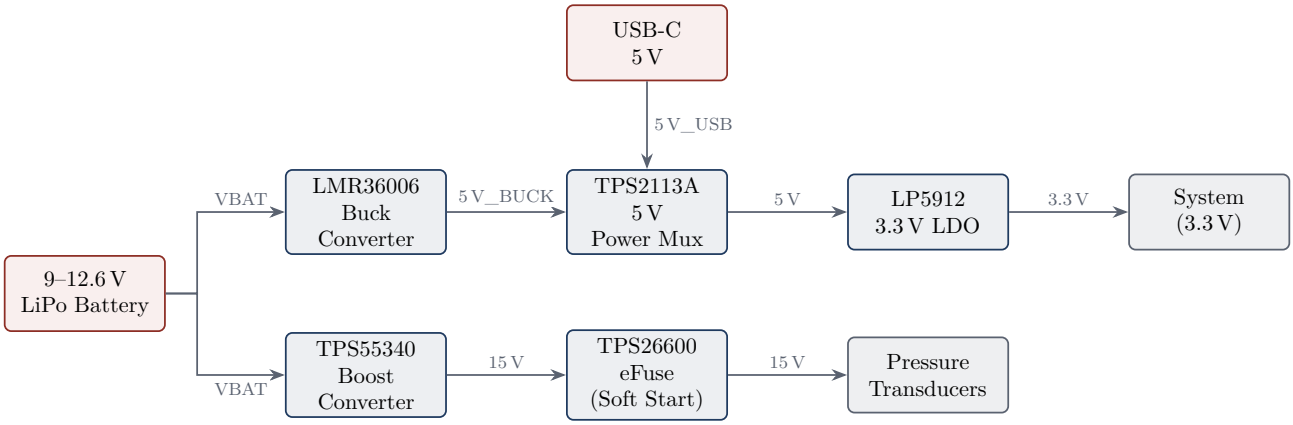


Figure 4: Power architecture.

The eFuse IC comes with adjustable soft-start capabilities, in which I set the soft-start time to 13ms. The 0.1u capacitor also sets the under-voltage lockout turn on delay to about 1.7ms, so this 100us spike never reaches the output during battery hot plugging. The 13ms soft start adds even more protection.

I decided to place a power-muxer IC with the inputs as the usb-c 5V and buck 5V, and the higher voltage is fed into the LDO to take advantage of its high PSRR and filter buck rail ripples, supplying a clean 3.3V source for the MCU and thermocouple amplifiers. This muxing also prevents backfeeding into the buck and usb-c.

3.2 Sensor Front Ends

3.2.1 Thermocouple Inputs

I used the MAX31856 thermocouple amplifier ICs, which have a wide range of thermocouple type compatibilities. Since the sampling rates of these are capped at about 11 Hz, I am exploring the design of our own system where we measure cold-junction temperature with a separate IC, amplify the millivolt difference between thermocouple wires, and create our own cold junction compensation algorithm. This would allow us to sample at higher rates in the future if needed.

3.2.2 Pressure Transducer Inputs

Although the power path covers our previous problem of voltage spikes, it is generally good practice to add protection circuitry at MCU pins that are broken out to the physical world to protect from accidental incorrect wiring and ESD events.

If there is a need for more pressure transducer inputs in the future, I am looking into ADC sampling ICs, which communicate via SPI.

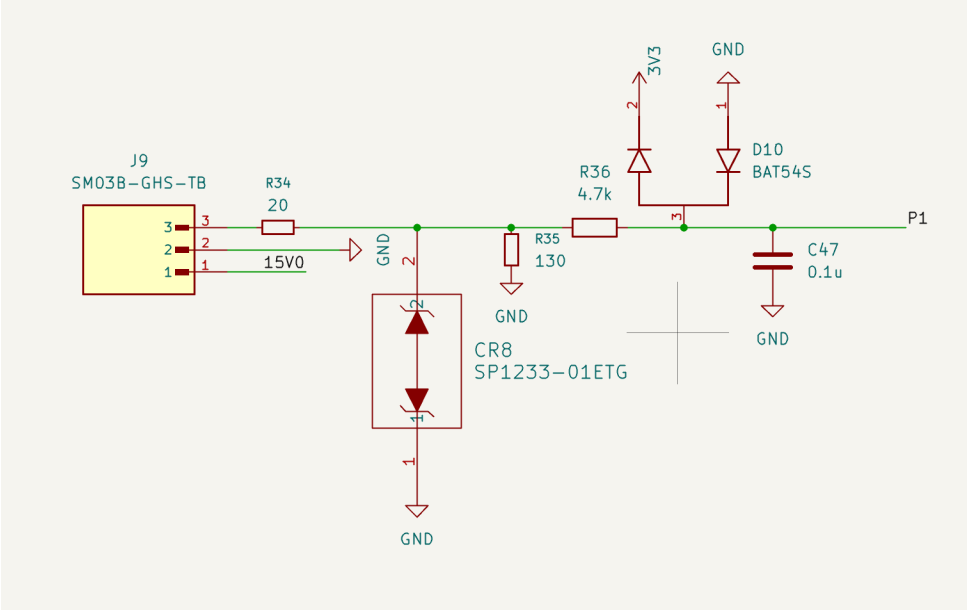


Figure 5: STM32 ADC-transducer protection circuitry

3.3 Communication Buses

I have broken out various communication lines, listed in Table 1, to allow for attaching other microcontrollers, radio transceiver breakout boards, and any other sensors needed on future rockets. The external SPI ADCs and custom thermocouple system can be implemented via these communication lines. I implemented a 4-to-16 decoder to generate SPI chip select pins while only using 4 MCU GPIO pins, allowing for the addition of 11 more SPI peripherals.

4 Next Steps

- Test PCB, looking for quality and reliability issues
- Write and optimize firmware for the fastest sampling rate
- Test board system, adding other breakouts/boards
- Design rocket flight telemetry board

5 PCB Layout

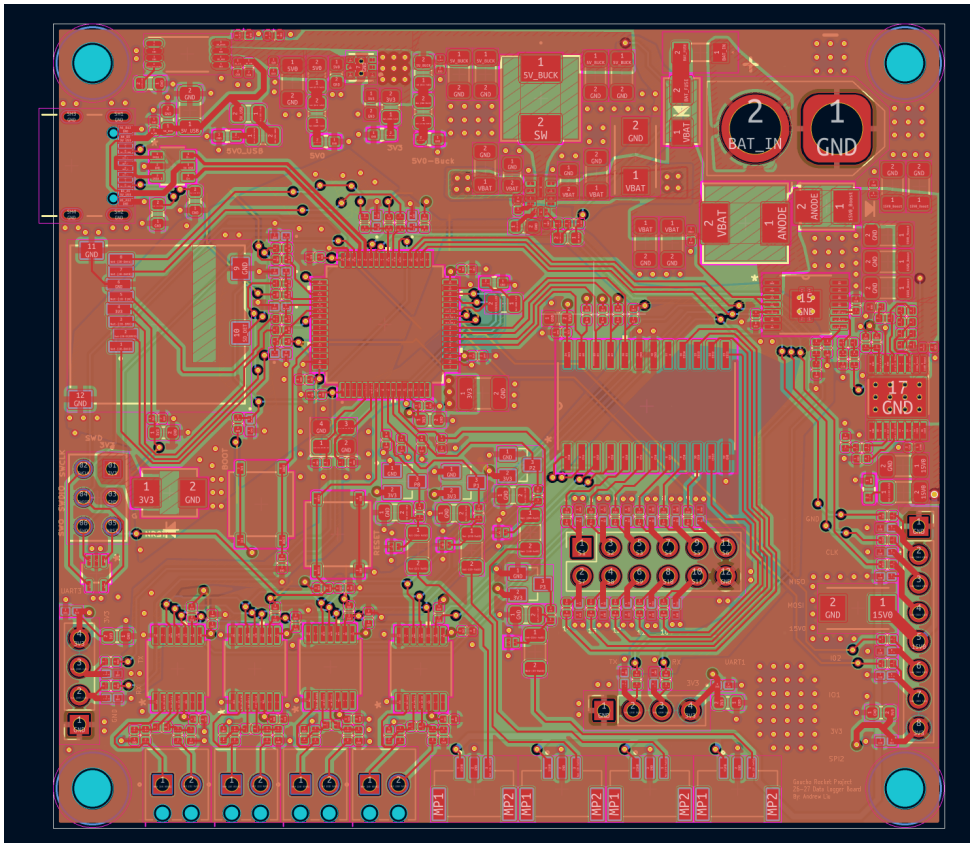
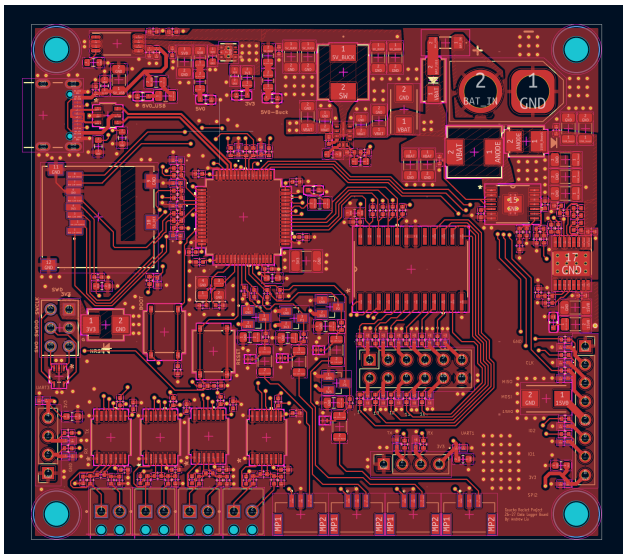
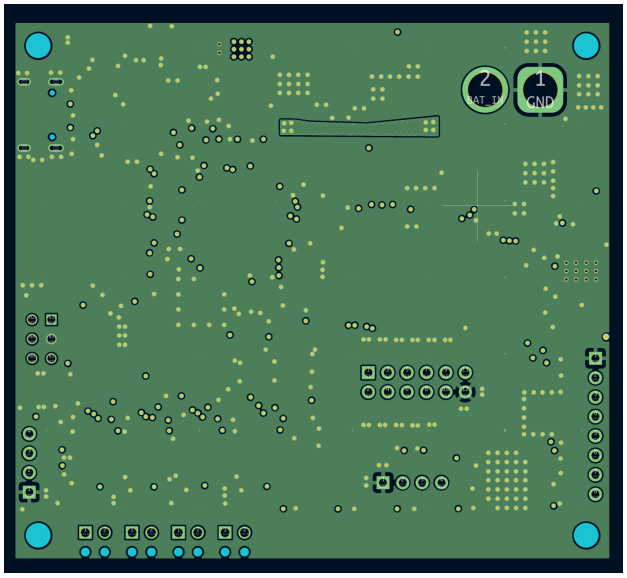


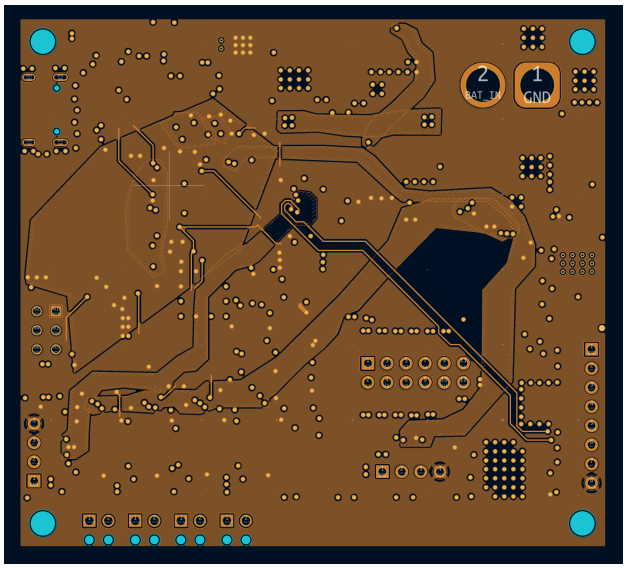
Figure 6: Board with all layers



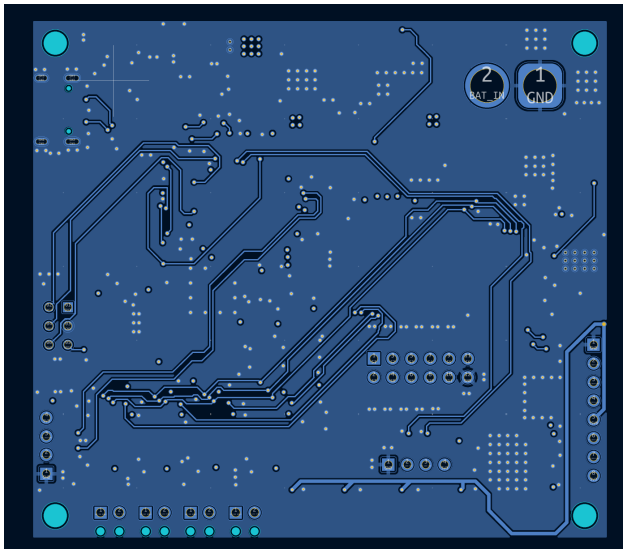
Layer 1 - Main component routing



Layer 2 - Ground fill



Layer 3 - 3.3V and ground split fill



Layer 4 - Extra routing and ground fill

Figure 7: Individual board layers

6 Schematic

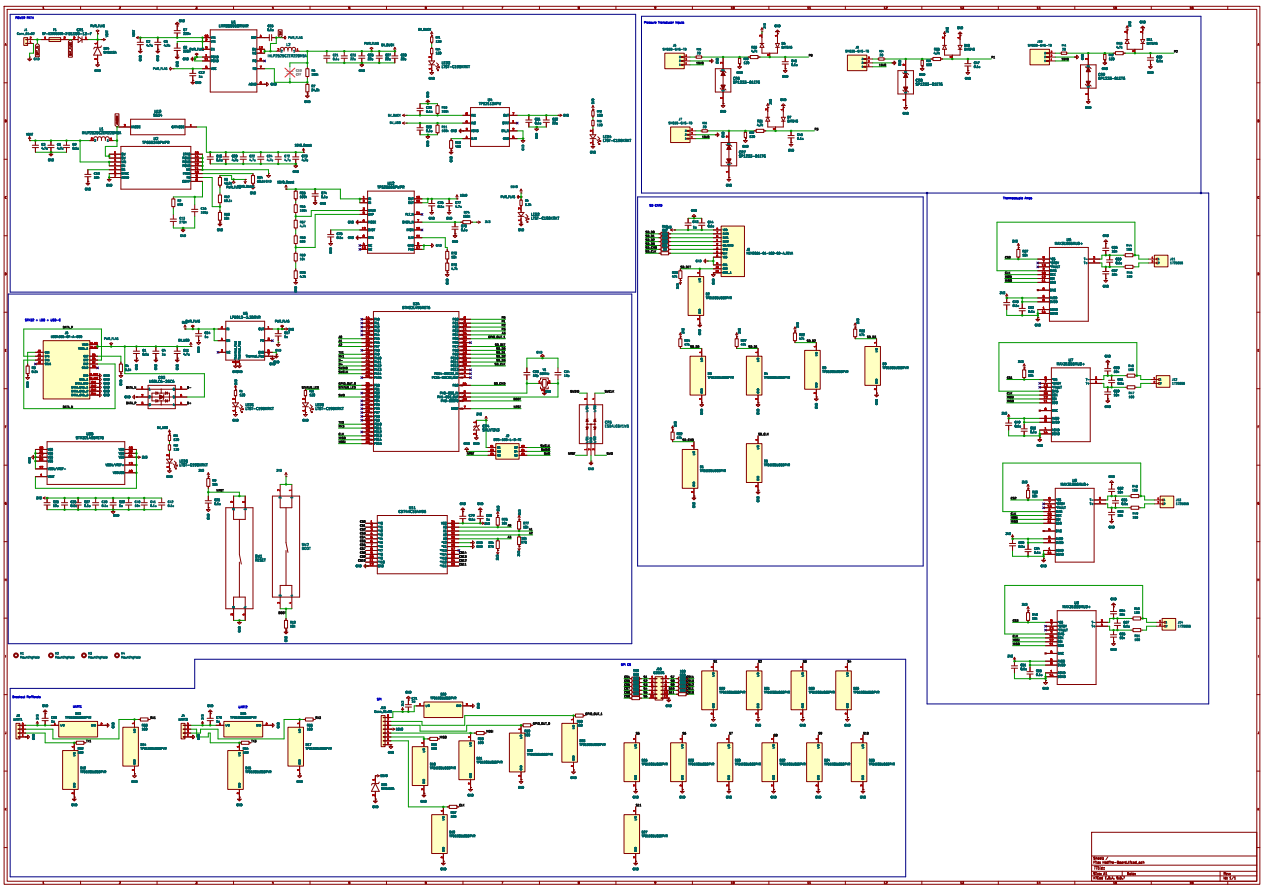


Figure 8: Schematic